Chapter 08 Part 1: Heat and Temperature

1. A 1.0 g sample of a cashew was burned in a calorimeter containing 1000. g of water, and the temperature of the water changed from 20.0°C to 25.0°C. In another experiment, a 3.0 g sample of a marshmallow was burned in a calorimeter containing 2000. g of water, and the temperature of the water changed from 25.0°C to 30.0°C. Based on the data, which of the following can be concluded about the energy content for 1.0 g of each of the two substances? (The specific heat of water is 4.2 J/(g·°C).)

(A) The combustion of 1.0 g of cashew releases less energy than the combustion of 1.0 g of marshmallow.

(B) The combustion of 1.0 g of cashew releases the same amount of energy as the combustion of 1.0 g of marshmallow.

(C) The combustion of 1.0 g of cashew releases more energy than the combustion of 1.0 g of marshmallow. ✓

(D) No comparison can be made because the two systems started with different masses of food, different masses of water, and different initial temperatures.

2. A 10. g cube of copper at a temperature T_1 is placed in an insulated cup containing 10. g of water at a temperature T_2 . If $T_1 > T_2$, which of the following is true of the system when it has attained thermal equilibrium? (The specific heat of copper is 0.385 J/(g·°C) and the specific heat of water is 4.18 J/(g·°C).)

(A) The temperature of the copper changed more than the temperature of the water. ✓

(B) The temperature of the water changed more than the temperature of the copper.

(C) The temperature of the water and the copper changed by the same amount.

(D) The relative temperature changes of the copper and the water cannot be determined without knowing T_1 and T_2 .

3. A 100g sample of a metal was heated to 100^{o}C and then quickly transferred to an insulated container holding 100g of water at 22^{o}C. The temperature of the water rose to reach a final temperature of 35^{o}C. Which of the following can be concluded?

(A) The metal temperature changed more than the water temperature did; therefore the metal lost more thermal energy than the water gained.

(B) The metal temperature changed more than the water temperature did, but the metal lost the same amount of thermal energy as the water gained. ✓

(C) The metal temperature changed more than the water temperature did; therefore the heat capacity of the metal must be greater than the heat capacity of the water.

(D) The final temperature is less than the average starting temperature of the metal and the water; therefore the total energy of the metal and water decreased.

4. A 50.0 g sample of Fe at 100°C is added to 500.0 mL of water at 35°C in a perfectly insulated container. Which of the following statements is true?

(A) The temperature of the water will decrease.

(B) The temperature change of the Fe will have the same magnitude as the temperature change of the water.

(C) The heat energy lost by the Fe will be less than the heat energy gained by the water.

(D) The heat energy lost by the Fe will be equal to the heat energy gained by the water. ✓

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5. A hot iron ball is dropped into a 200. g sample of water initially at 50°C. If 8.4 kJ of heat is transferred from the ball to the water, what is the final temperature of the water? (The specific heat of water is 4.2 J/(g·°C).)

(A) 40°C

(B) 51°C

(C) 60°C ✓

(D) 70°C

6. A quantity of $10{,}000\ \mathrm{J}$ of thermal energy is transferred to four different substances. In which of the following experiments could the greatest mass of the given substance undergo the indicated change?

(A) Melting ice at $0.^\circ\mathrm{C}$ to form liquid water at $0.^\circ\mathrm{C}$ $(\Delta H_{fus} = 330\ \mathrm{J/g})$

(B) Boiling liquid acetone at $56^\circ\mathrm{C}$ to form acetone vapor at $56^\circ\mathrm{C}$ $(\Delta H_{vap} = 540\ \mathrm{J/g})$

(C) Increasing the temperature of liquid glycerin from $20.^\circ\mathrm{C}$ to $120.^\circ\mathrm{C}$ $(\mathrm{Specific\ \ heat} = 2.4\ \mathrm{J/(g\cdot{}^\circ C)})$

(D) Increasing the temperature of solid copper from $0.^\circ\mathrm{C}$ to $200.^\circ\mathrm{C}$ $(\mathrm{Specific\ \ heat} = 0.40\ \mathrm{J/(g\cdot{}^\circ C)})$ ✓

7. For a classroom demonstration, a chemistry teacher puts samples of two different pure solid powders in a beaker. The teacher places the beaker on a small wooden board with a wet surface, then stirs the contents of the beaker. After a short time the students observe that the bottom of the beaker is frozen to the wood surface. The teacher asks the students to make a claim about the observation and to justify their claims. Which of the following is the best claim and justification based on the students' observation?

(A) An exothermic chemical change occurred because heat flowed from the contents of the beaker to the room.

(B) An exothermic physical change occurred because heat flowed from the contents of the beaker and the water on the board to the room.

(C) An endothermic physical change occurred because the freezing of water is an endothermic process.

(D) An endothermic chemical change occurred because the temperature of the beaker and the water on the board decreased as heat was absorbed by the reaction. ✓

8. A $1.0\ \mathrm{mol}$ sample of $\mathrm{He}(g)$ at $25^\circ\mathrm{C}$ is mixed with a $1.0\ \mathrm{mol}$ sample of $\mathrm{Xe}(g)$ at $50^\circ\mathrm{C}$. Which of the following correctly predicts the changes in average kinetic energy and the average speed of the $\mathrm{Xe}(g)$ atoms that will occur as the mixture approaches thermal equilibrium?

Chapter 08 Part 1: Heat and Temperature

(A)

Average Kinetic Energy of Xe Atoms	Average Speed of Xe Atoms
Will increase	Will increase

(B)

Average Kinetic Energy of Xe Atoms	Average Speed of Xe Atoms
Will increase	Will decrease

(C)

Average Kinetic Energy of Xe Atoms	Average Speed of Xe Atoms
Will decrease	Will increase

(D) ✓

Average Kinetic Energy of Xe Atoms	Average Speed of Xe Atoms
Will decrease	Will decrease

9.

Substance	Specific Heat $(J/(g \cdot °C))$	Mass of Sample (g)
$Fe(s)$	0.46	1.0
$Al(s)$	0.91	1.0

Samples of Fe and Al are heated from $25°C$ to $75°C$. Based on the information in the table above, the change in the average kinetic energy of the atoms in the samples is

(A) higher for the Fe sample, because with its lower specific heat it absorbs more energy

(B) higher for the Al sample, because with its higher specific heat it absorbs more energy

(C) the same for both metal samples, because the initial and final temperatures are the same ✓

(D) the same for both metal samples, because the masses are equal

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10. Which of the following is a list of the minimum amount of data needed for determining the molar enthalpy of solution of $\mathrm{KCl}(s)$ in pure $\mathrm{H_2O}(l)$? (Assume that the $\mathrm{KCl}(aq)$ has the same specific heat capacity as pure water and that the initial temperatures of the $\mathrm{KCl}(s)$ and the water are the same.)

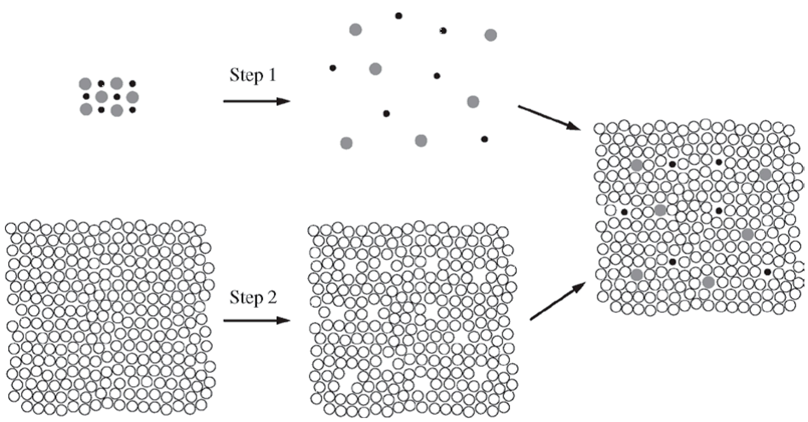
(A) Mass of $\mathrm{KCl}(s)$, initial temperature of the water, and final temperature of the solution

(B) Mass of $\mathrm{H_2O}$, initial temperature of the water, and final temperature of the solution

(C) Mass of $\mathrm{KCl}(s)$, mass of $\mathrm{H_2O}$, initial temperature of the water, and final temperature of the solution ✓

(D) Mass of $\mathrm{KCl}(s)$, mass of $\mathrm{H_2O}$, initial temperature of the water, final temperature of the solution, and atmospheric pressure

11.



The dissolution of an ionic solute in a polar solvent can be imagined as occurring in three steps, as shown in the figure above. In step 1, the separation between ions in the solute is greatly increased, just as will occur when the solute dissolves in the polar solvent. In step 2, the polar solvent is expanded to make spaces that the ions will occupy. In the last step, the ions are inserted into the spaces in the polar solvent. Which of the following best describes the enthalpy change, ΔH, for each step?

(A) All three steps are exothermic.

(B) All three steps are endothermic.

(C) Steps 1 and 2 are exothermic, and the final step is endothermic.

(D) Steps 1 and 2 are endothermic, and the final step is exothermic. ✓

12. A $30.$ g sample of $\mathrm{Al}(s)$ is heated to $50^\circ\mathrm{C}$ and placed in a calorimeter containing 150 g of water at $20^\circ\mathrm{C}$. As the system approaches thermal equilibrium, energy is transferred

(A) from the $\mathrm{Al}(s)$ to the water and the temperature of the water decreases

(B) from the $\mathrm{Al}(s)$ to the water and the temperature of the $\mathrm{Al}(s)$ decreases ✓

(C) from the water to the $\mathrm{Al}(s)$ and the temperature of the water increases

(D) from the water to the $\mathrm{Al}(s)$ and the temperature of the $\mathrm{Al}(s)$ increases

13. In an insulated cup of negligible heat capacity, 50. g of water at 40.°C is mixed with 30. g of water at 20.°C. The final temperature of the mixture is closest to

Chapter 08 Part 1: Heat and Temperature

(A) 22°C

(B) 27°C

(C) 30.°C

(D) 33°C ✓

(E) 38°C

14. When 5.0 g of $NH_4ClO_4(s)$ is added to 100. mL of water in a calorimeter, the temperature of the solution formed decreases by 3.0°C. If 5.0 g of $NH_4ClO_4(s)$ is added to 1000. mL of water in a calorimeter initially at 25.0°C, the final temperature of the solution will be approximately

(A) 22.0°C

(B) 24.7°C ✓

(C) 25.3°C

(D) 28.0°C

15. In the spring, blossoms on cherry trees can be damaged when temperatures fall below $-2^{\circ}\mathrm{C}$. When the forecast calls for air temperatures to be below $-5^{\circ}\mathrm{C}$ for a few hours one night, a farmer sprays his blossoming cherry trees with water, claiming that the blossoms will be protected by the water as it freezes. Which of the following is a correct scientific justification for spraying water on the blossoms to protect them from temperatures below $-2^{\circ}\mathrm{C}$?

(A) Water on the blossoms will not freeze unless the air temperature falls significantly below $-5^{\circ}\mathrm{C}$.

(B) Water is a good thermal conductor that transfers heat from the cold air to the blossoms, keeping the blossoms from going below $-2^{\circ}\mathrm{C}$.

(C) The freezing of water is an endothermic process; thus, water that freezes on the blossoms absorbs heat from the atmosphere, which in turn keeps the blossoms above $0^{\circ}\mathrm{C}$.

(D) The freezing of water is an exothermic process; thus, water that freezes on the blossoms releases heat to keep the blossoms at or above $-2^{\circ}\mathrm{C}$. ✓

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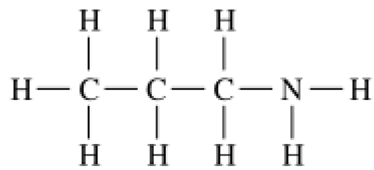
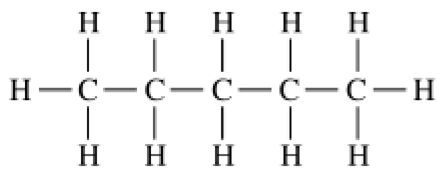
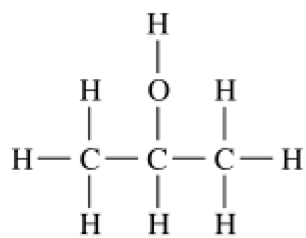
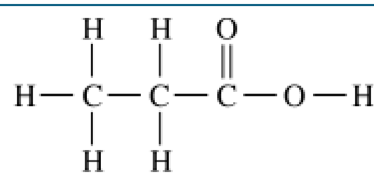
Substance	Structural Formula	Molar Mass (g/ mol)
Propylamine		59
Pentane		72
2-propanol		60
Propanoic acid		74

A student was given samples of four different unknown liquids. The unknown liquids are propylamine, pentane, 2-propanol, and propanoic acid. The structures and molar masses of the compounds are shown in the table above. On the basis of this information, the student designed some experiments to identify the samples.

16. Which of the following statements correctly identifies which compound, pentane or propanoic acid, has the higher boiling point and provides a valid justification?

(A) Pentane, because it has greater London dispersion forces than propanoic acid does

(B) Pentane, because it can form more hydrogen bonds per molecule than propanoic acid can

(C) Propanoic acid, because double bonds are harder to break than single bonds

(D) Propanoic acid, because it forms hydrogen bonds, whereas pentane does not

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17. In an experiment to estimate the enthalpy change of a reaction, a student makes two aqueous solutions, each containing one of the reactants. The student combines the solutions, both originally at the same temperature, in a calorimeter and records the final temperature of the mixture. In addition to the masses of the solutions and the temperature change of the mixture, which of the following pieces of information does the student need to calculate the enthalpy change of the reaction?

(A) The density of the reaction mixture

(B) The specific heat capacity of the reaction mixture

(C) The boiling point of the reaction mixture

(D) The heat of fusion of the reaction mixture

18.

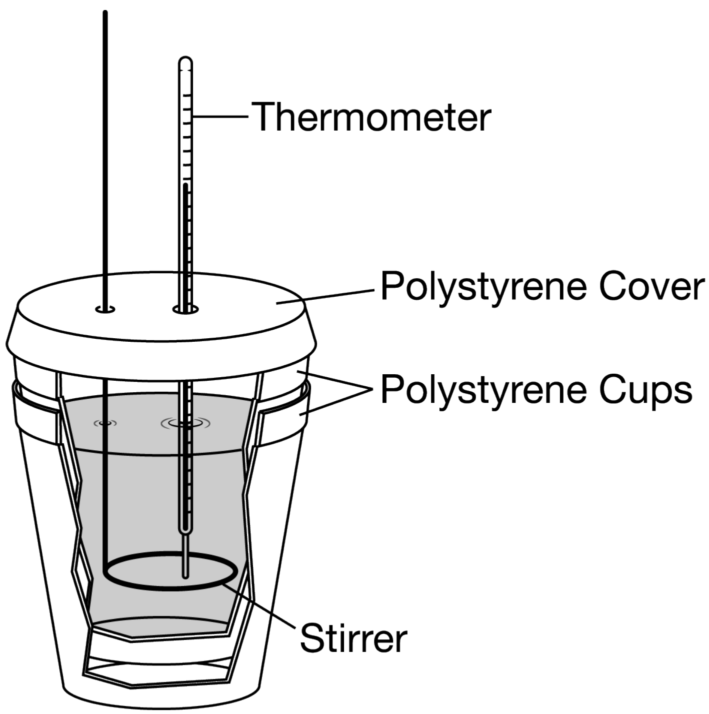

Diagram 1

For an experiment, $50.0\ \mathrm{g}$ of H_2O was added to a coffee-cup calorimeter, as shown in the diagram above. The initial temperature of the H_2O was $22.0^\circ\mathrm{C}$, and it absorbed $300.\ \mathrm{J}$ of heat from an object that was carefully placed inside the calorimeter. Assuming no heat is transferred to the surroundings, which of the following was the approximate temperature of the H_2O after thermal equilibrium was reached? Assume that the specific heat capacity of H_2O is $4.2\ \mathrm{J/(g \cdot K)}$.

(A) $21.3^\circ\mathrm{C}$

(B) $22.0^\circ\mathrm{C}$

(C) $22.7^\circ\mathrm{C}$

(D) $23.4^\circ\mathrm{C}$

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19. A piece of $Fe(s)$ at $25^{\circ}C$ is placed into $H_2O(l)$ at $75^{\circ}C$ in an insulated container. A student predicts that when thermal equilibrium is reached, the Fe atoms, being more massive than the H_2O molecules, will have a higher average kinetic energy than the H_2O molecules. Which of the following best explains why the student's prediction is incorrect?

(A) At thermal equilibrium, the less massive H_2O molecules would have a higher average kinetic energy than the Fe atoms because they are more free to move than are the Fe atoms.

(B) At thermal equilibrium, the collisions between the Fe atoms and the H_2O molecules would cease because the average kinetic energies of their particles would have become the same.

(C) At thermal equilibrium, the movement of both the Fe atoms and the H_2O molecules would cease; thus, the average kinetic energy of their particles would have to be the same.

(D) At thermal equilibrium, the average kinetic energy of the Fe atoms cannot be greater than that of the H_2O molecules; the average kinetic energies must be the same according to the definition of thermal equilibrium. ✓

$$K(s) + \tfrac{1}{2}Cl_2(g) \rightarrow KCl(s) \qquad \Delta H° = -437 \text{kJ/mol}_{rxn}$$

The elements K and Cl react directly to form the compound KCl according to the equation above. Refer to the information above and the table below to answer the questions that follow.

Process	$\Delta H°$ (kJ/mol$_{rxn}$)
$K(s) \rightarrow K(g)$	v
$K(g) \rightarrow K^+(g) + e^-$	w
$Cl_2(g) \rightarrow 2\,Cl(g)$	x
$Cl(g) + e^- \rightarrow Cl^-(g)$	y
$K^+(g) + Cl^-(g) \rightarrow KCl(s)$	z

20. Which of the values of $\Delta H°$ for a process in the table is (are) less than zero (i.e., indicate(s) an exothermic process)?

(A) z only

(B) y and z only ✓

(C) x, y, and z only

(D) w, x, y, and z

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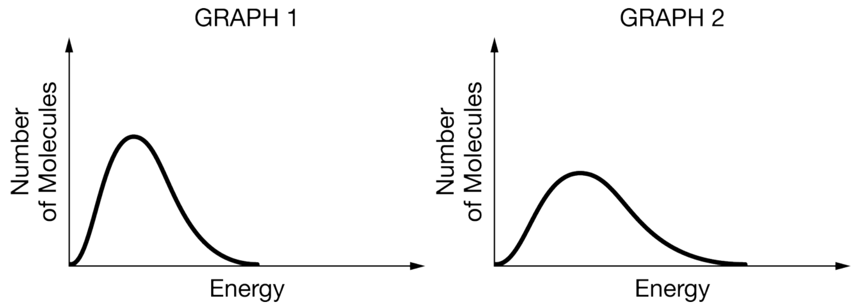
21.



The graphs above show Maxwell-Boltzmann distributions for one-mole samples of $\mathrm{Ar}(g)$. Graph 1 shows the distribution of particle energies at $300\ \mathrm{K}$ and graph 2 shows the distribution of particle energies at $600\ \mathrm{K}$. A student predicts that if the samples are combined in an insulated container and thermal equilibrium is attained, then the most probable particle energy will be between the most probable energy shown in graph 1 and the most probable energy shown in graph 2. Which of the following is the best justification for the student's claim?

(A) When the samples are combined, the gas particles will collide with one another, with the net effect being that the speed of the lowest energy particles decreases while the speed of the highest energy particles increases, leaving the average speed of the particles in the original samples unchanged.

(B) When the samples are combined, the gas particles from each sample will collide with the gas particles from the other sample until every particle in the mixture has the same speed, which is between the average speed of the particles in the hotter sample and the average speed of the particles in the cooler sample.

(C) When the samples are combined, the gas particles collide with one another until every particle in the mixture has the same kinetic energy, which is between the average kinetic energy of the particles in the hotter sample and the average kinetic energy of the particles in the cooler sample.

(D) When the samples are combined, the gas particles will collide with one another, with the net effect being that energy will be transferred from the more energetic particles to the less energetic particles until a new distribution of energies is achieved at a temperature between $300\ \mathrm{K}$ and $600\ \mathrm{K}$. ✓

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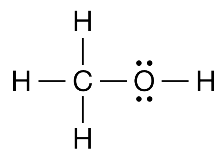
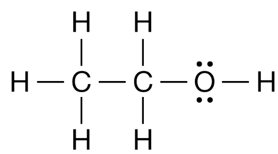
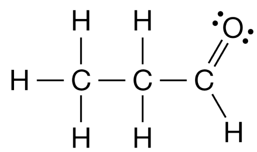
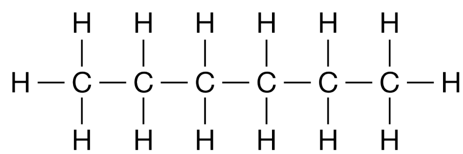
Compound	Structure	Boiling Point (°C)
Methanol		65
Ethanol		78
Propanal		49
Hexane		69

22. When a $50.\ mL$ sample of C_2H_5OH is mixed with a $50.\ mL$ sample of H_2O, the resulting mixture has a volume of $95\ mL$, and the container is warm to the touch. Which of the following best describes these observations?

(A) A chemical reaction occurs, as evidenced by the volume of the resultant mixture being less than the total volume of the initial components.

(B) A chemical change occurs, as evidenced by the formation of new covalent bonds releasing more energy than is absorbed by the breaking of the existing covalent bonds.

(C) A physical change occurs, and the solvation process is exothermic.

(D) A physical change occurs, and the solvation process is endothermic.

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23.

Mass of water	50.003 g
Temperature of water	$24.95^{\circ}C$
Specific heat capacity for water	$4.184\ J/g^{\circ}C$
Mass of metal	63.546 g
Temperature of metal	$99.95^{\circ}C$
Specific heat capacity for metal	?
Final temperature	$32.80^{\circ}C$

In an experiment to determine the specific heat of a metal, a student transferred a sample of the metal that was heated in boiling water into room-temperature water in an insulated cup. The student recorded the temperature of the water after thermal equilibrium was reached. The data are shown in the table above. Based on the data, what is the calculated heat q absorbed by the water reported with the appropriate number of significant figures?

(A) 1600 J

(B) 1640 J

(C) 1642 J

(D) 1642.3 J

24.

Cup	Material of Spoon	Initial Temperature of Spoon ($^{\circ}C$)	Mass of Spoon (g)	Specific Heat Capacity ($J/g\ ^{\circ}C$)
A	Aluminum	20	10.0	0.90
B	Ceramic	20	10.0	0.80
C	Steel	20	20.0	0.45
D	Silver	20	40.0	0.23

Four identical 50 mL cups of coffee, originally at $95^{\circ}C$, were stirred with four different spoons, as listed in the table above. In which cup will the temperature of the coffee be highest at thermal equilibrium? (Assume that the heat lost to the surroundings is negligible.)

(A) Cup A

(B) Cup B

(C) Cup C

(D) Cup D

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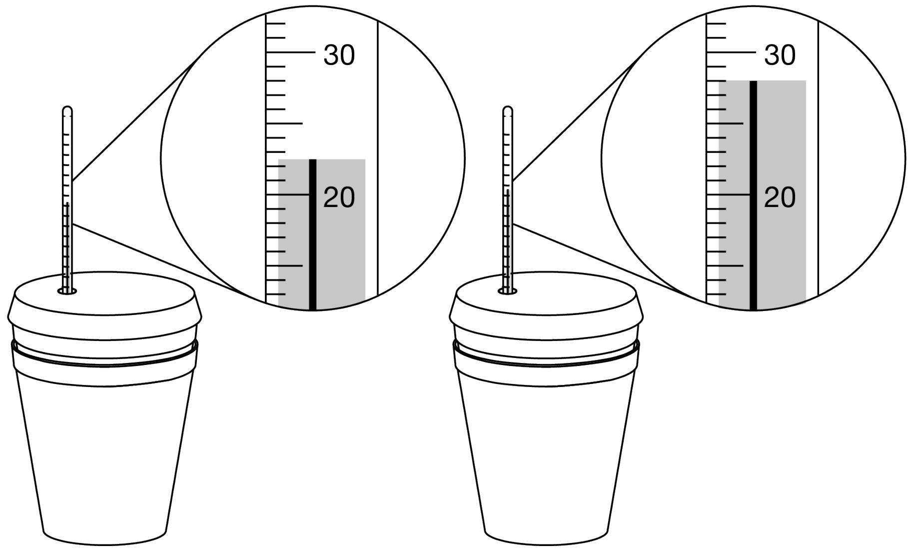
25.



A student mixes 50 mL of $1.0 \ M \ \text{HCl}$ and 50 mL of $1.0 \ M \ \text{NaOH}$ in a coffee-cup calorimeter and observes the change in temperature until the mixture reaches thermal equilibrium. The initial and final temperatures $(^\circ\text{C})$ of the mixture are shown in the diagram above of the laboratory setup.

Based on the results, what is the change in temperature reported with the correct number of significant figures?

(A) 5.5°C ✓

(B) 5.50°C

(C) 5.800°C

(D) 6°C

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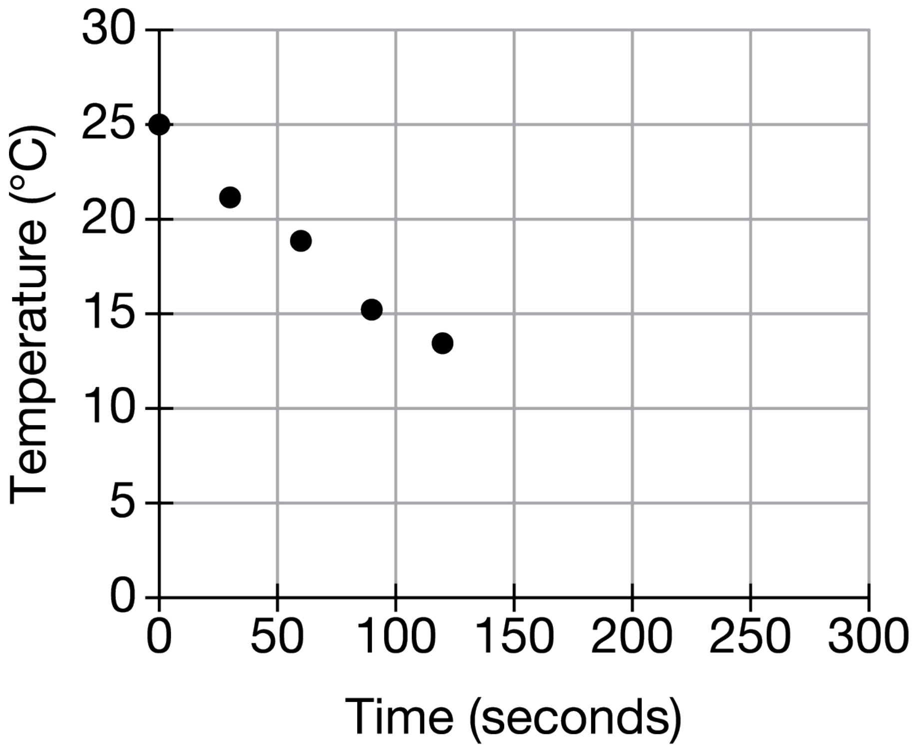
26.



A student adds 50.0 g of liquid water at $25.0^{\circ}\mathrm{C}$ to an insulated container fitted with a temperature probe. The student then adds 10.0 g of ice at $0.0^{\circ}\mathrm{C}$, closes the container, and measures the temperature at different intervals. Part of the data is shown in the graph above. The student predicts that the temperature will continue to decrease then level out to a constant temperature. Which of the following best explains why the student's prediction is correct?

(A) The H_2O molecules initially in the ice and the molecules initially in the liquid will have the same average kinetic energy. ✓

(B) The transfer of energy between the H_2O molecules in the ice and liquid water stops once all the molecules are in the liquid phase.

(C) Once all of the H_2O molecules are in the liquid phase, the individual molecular speeds either increase or decrease until all the particles have the same speed.

(D) Once all of the H_2O molecules are in the liquid phase, collisions between them virtually stop as they reach an equilibrium distance from their neighboring molecules.

27.

$$\mathrm{NaOH}(aq) + \mathrm{HCl}(aq) \rightarrow \mathrm{H_2O}(l) + \mathrm{NaCl}(aq)$$

A student is trying to determine the heat of reaction for the acid-base neutralization reaction represented above. The student uses 0.50 *M* NaOH and 0.50 *M* HCl solutions. Which of the following situations, by itself, would most likely result in the LEAST error in the calculated value of the heat of reaction?

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(A) The thermometer was incorrectly calibrated and read 0.5 Celsius degree too high during the procedure.

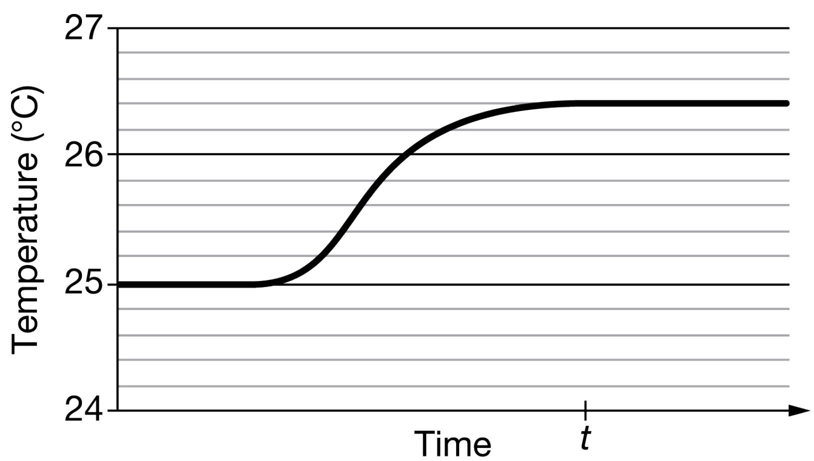
(B) The volume of the acid solution added to the calorimeter was actually 1.0 mL less than what was recorded.

(C) The calorimeter was poorly insulated, and some heat escaped to the atmosphere during the procedure.

(D) The actual molarity of the base solution was 0.53 *M* but was recorded as 0.50 *M*.

(E) The final temperature of the mixture was taken before the contents of the calorimeter had reached thermal equilibrium.

28. For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.



A student did an experiment to determine the specific heat capacity of a metal alloy. The student put a sample of the alloy in boiling water for several minutes, then quickly transferred the alloy into a calorimeter containing water originally at $25^\circ C$. The temperature of the water was monitored over time. The data are given in the graph above.

(a) What is the value of ΔT that the student should use to calculate the value of q, the heat gained by the water?

(b) In terms of what occurs at the particulate level, explain how the temperature of the water increases after the alloy sample is added.

(c) The student claims that thermal equilibrium is reached at time t. Justify the student's claim. In your justification, include a description of what occurs at the particulate level when the alloy and the water have reached thermal equilibrium.

Part (a)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

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0	1

The response indicates that ΔT is $1.4^\circ\mathrm{C}$.

Part (b)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1

The response indicates that the atoms in the alloy transfer kinetic energy to the molecules in the water.

Part (c)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2

The response indicates both of the following criteria:

- ☐ The claim is correct because the temperature levels off at that time.
- ☐ At equilibrium the average kinetic energy of the atoms in the alloy is equal to the average kinetic energy of the water molecules.